# AI now helps write the software that runs machines. No mainstream language can prove what that software will never do.

Batteries, flywheels and flight controls run on code that people test and review, and then trust.

**Mentl is a programming language whose compiler proves it.**

# Mentl proves what software can never do.

You write a rule into the code, such as "this never touches the internet" or "this never writes a file".

The compiler checks the rule through every function the code calls. **Break it, and the program won't build.**

Seven months and 2,600 commits since March. The compiler is written in Mentl and compiles itself. Open source: github.com/ampactor-labs/mentl

✕ Refused

**One line broke the rule.**

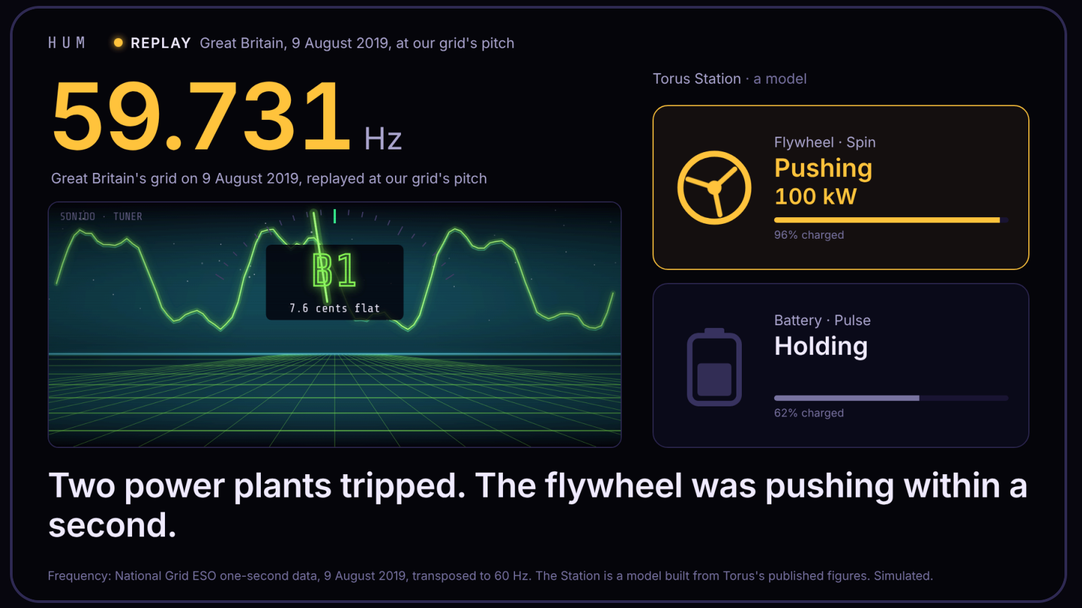
```
let price = price_m("https://api.eia.gov/…")
```

This decision is marked as never reaching the internet. Add one line that reads a price online, and the compiler won't build it. Its error names the path the price lookup took.

Show it working

# Built this week: software that measures the power grid from a wall's wiring.

- **A guitar pickup picks up the grid's hum** in any room, and my audio library, sonido, measures its frequency to a thousandth of a hertz.
- **A model of a Torus Station decides** what to do, typically in under a millisecond, with software written in Mentl.
- **That software is proven** unable to reach the internet while it decides.




Before Monday 4 PM: the language, 2,610 commits since March 15, and sonido, 461 commits since January 25. Since then: everything on this slide.

# Who needs it in Utah

**TORUS · SALT LAKE CITY**

Builds flywheel-and-battery Stations that steady the grid. The software inside each Station is where a proof would matter first. Torus is judging this demo.

**HILL AFB · 309TH SOFTWARE ENGINEERING GROUP**

Maintains flight software for the F-16, the F-35 and the missile fleet, where every claim about the software needs evidence.

First conversation I'm asking for: Torus engineering, this month.
First hires: a controls engineer and a compiler engineer, in Salt Lake City.

What $10,000 unlocks

# $10,000 puts this controller beside a real battery in Utah within ninety days.

**Day 30** Form the Utah company. Run the controller on a battery inverter on my bench.

**Day 60** Run it beside a real battery at one Utah site.

**Day 90** A pilot agreement with that site, and an SBIR proposal with Nucleus Grow.

**$7,000** for the inverter and battery bench. **$3,000** to form the company, register for federal grants and install at the site.

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**Three asks.** A Utah operator who will host it. An introduction to Hill AFB's 309th. A co-founder who knows power electronics.

BACKUP

# What is proven, exactly

| THE CONTROLLER'S RULE | STATUS |
| --- | --- |
| No network, file, clock or other call outside the program while it decides | Proven by the compiler through every call |
| No memory allocated while it decides | Proven |
| The flywheel within 100 kW, the battery within 300 kW | Proven: each machine answers a command with a power inside its rating, and the compiler checks the answer |
| It never stops with an error | Proven for every function the controller defines; one byte-range check inside the library's print routine is not proven yet |
| Its decisions are good | Tested on recordings, not proven |

The compiler is written in Mentl, and the pinned binary rebuilds itself byte for byte. The first four rows are checked through every call, and a copy that breaks one does not build.

Checked by the compiler: mentl audit examples/grid/station.mn, and mentl why for each rule

BACKUP

# From the laptop to the pedal: the day-30 engineering

| WHAT STANDS IN THE WAY | MEASURED TODAY | THE FIX |
|---|---|---|
| No WebAssembly engine on the pedal's chip | The controller is a 37 KB WebAssembly file | Embed an interpreter written in Rust, or compile the file to C |
| Built for threads | Shared memory and atomic instructions | A single-threaded build; the controller uses no threads |
| Memory | Fixed: the same peak after 300,000 readings as after one | Done; each reading runs in an arena that is reclaimed when it ends |
| Speed | About 291,000 instructions per reading | Pass numbers instead of text |

The decision allocates nothing and the loop reclaims each reading's memory. The work left is the engine on the chip.